

Ex. No: 7 Banker's Algorithm

PROGRAM: BANKER'S ALGORITHM

C PROGRAM

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
int n, m, i, j, k;
```

```
int allocation[10][10], max[10][10];
```

```
int need[10][10];
```

```
int available[10];
```

```
int finish[10] = {0};
```

```
int safeSeq[10];
```

```
int count = 0;
```

```
printf("Enter Number of Processes: ");
```

```
scanf("%d", &n);
```

```
printf("Enter Number of Resources: ");
```

```
scanf("%d", &m);
```

```
printf("
```

```
\nEnter Allocation Matrix:
```

```
\n");
```

```
for(i = 0; i < n; i++)
```

```
for(j = 0; j < m; j++)
```

```
scanf("%d", &allocation[i][j]);
```

```
printf("
```

```
\nEnter Maximum Matrix:
```

```
\n");
```

```
for(i = 0; i < n; i++)
```

```
for(j = 0; j < m; j++)
```

```
scanf("%d", &max[i][j]);
```

```
printf("\nEnter Available Resources:\n");
```

```
for(i = 0; i < m; i++)  
scanf("%d", &available[i]);  
for(i = 0; i < n; i++)  
{  
for(j = 0; j < m; j++)  
{  
need[i][j] = max[i][j]
```

```
- allocation[i][j];
```

```
}  
}  
while(count < n)  
{  
int found = 0;  
for(i = 0; i < n; i++)  
{  
if(finish[i] == 0)  
{  
for(j = 0; j < m; j++)  
{  
if(need[i][j] > available[j])  
break;  
}  
if(j == m)  
{  
for(k = 0; k < m; k++)
```

```
available[k] += allocation[i][k];
safeSeq[count++] = i;
finish[i] = 1;
found = 1;
}
}
}
if(found == 0)
{
printf("\nSystem is NOT in Safe State\n");
return 0;
}
}
printf("\nSystem is in Safe State\n");
printf("Safe Sequence: ");
for(i = 0; i < n; i++)
printf("P%d ", safeSeq[i]);
printf("\n");
return 0;
}
```

Output:

```
(kali㉿kali)-[~]
$ ./banker
Enter Number of Processes: 2
Enter Number of Resources: 2
Enter Allocation Matrix:
2
34
4
5
Enter Maximum Matrix:
5
4
3
5
Enter Available Resources:
6
5
System is in Safe State
Safe Sequence: P0 P1
```

SHELL SCRIPT

```
#!/bin/bash

echo "Enter Number of Processes:"
read n

echo "Enter Safe Sequence (space separated):"
read -a seq

echo "Safe Sequence is:"

for ((i=0;i<n;i++))
do
echo -n "P${seq[i]}"
done

echo

echo "System is in Safe State"
```

Output:

```
(kali㉿kali)-[~]
└─$ chmod +x banker.sh
(kali㉿kali)-[~]
└─$ ./banker.sh
Enter Number of Processes: greater.sh
1
Enter Safe Sequence (space separated):
2
Safe Sequence is: 4 2
P2
System is in Safe State
(kali㉿kali)-[~]
```

deadlock processes:

Then:

Enter S:

Enter:

1 3 4 0

Ask anyt